NCERT Class 11 Biology — Chapter 22

Chemical Coordination and Integration — Comprehensive NEET Line-by-Line Notes

100% NCERT Coverage | Detailed Theory & Disorders | Mechanism of Action | NEET PYQ Based Sheet

1. INTRODUCTION & ENDOCRINE SYSTEM DEFINITION

In human body, neural system and endocrine system jointly coordinate and integrate all activities of organs. Neural system provides point-to-point rapid coordination (short-lived), whereas endocrine system provides chemical integration via hormones (slow & long-lasting).

- **Endocrine Glands:** Ductless glands that pour secretions directly into extracellular fluid / blood.
- **Classical Definition of Hormone:** A chemical produced by endocrine glands and released into blood and transported to a distantly located target organ.
- **Current Scientific Definition:** Hormones are **non-nutrient chemicals** which act as **intercellular messengers** and are produced in **trace amounts**.
- Invertebrates possess very simple endocrine systems with few hormones, whereas vertebrates have a large number of chemicals acting as hormones.

2. HUMAN ENDOCRINE SYSTEM

The endocrine glands and hormone-producing diffused tissues/cells located in different parts of our body constitute the endocrine system. Major endocrine glands: **Hypothalamus, Pituitary, Pineal, Thyroid, Parathyroid, Thymus, Adrenal, Pancreas, Gonads (Testis & Ovary).**

3. HYPOTHALAMUS

Located at the base of **diencephalon (forebrain)**. Contains several groups of neurosecretory cells called **nuclei** which produce hormones that regulate synthesis and secretion of pituitary hormones.

- **Releasing Hormones:** Stimulate pituitary secretion. E.g., **GnRH (Gonadotropin Releasing Hormone)** stimulates anterior pituitary to release gonadotropins (LH & FSH).
- **Inhibiting Hormones:** Inhibit pituitary secretion. E.g., **Somatostatin** inhibits release of Growth Hormone from pituitary.
- **Portal Circulation Connection:** Hormones reaching anterior pituitary pass through **hypophyseal portal system**.
- **Direct Neuro-regulation:** Posterior pituitary is under direct neural control of hypothalamus. Oxytocin and Vasopressin are synthesized in hypothalamus and transported axonally to posterior pituitary.

4. PITUITARY GLAND (HYPOPHYSIS)

Located in a bony cavity of sphenoid bone called **Sella Turcica** and attached to hypothalamus by an **Infundibulum** stalk.

Anatomical Division:

1. **Adenohypophysis:**
   - **Pars Distalis (Anterior Pituitary):** Secretes GH, PRL, TSH, ACTH, LH, FSH.
   - **Pars Intermedia:** In humans, almost merged with Pars Distalis. Secretes **MSH (Melanocyte Stimulating Hormone)**.
2. **Neurohypophysis:**
   - **Pars Nervosa (Posterior Pituitary):** Stores and releases hormones synthesized by hypothalamus: **Oxytocin** and **Vasopressin (ADH)**.

Hormones of Pituitary & Functions:

- **GH (Growth Hormone):** Over-secretion in childhood $ ightarrow$ **Gigantism**; Under-secretion in childhood $ ightarrow$ **Pituitary Dwarfism**; Hypersecretion in adults (middle age) $ ightarrow$ **Acromegaly** (severe disfigurement of face, premature death if unchecked).
- **PRL (Prolactin):** Growth of mammary glands and formation of milk in them.
- **TSH (Thyroid Stimulating Hormone):** Stimulates synthesis & secretion of thyroid hormones (T_3, T_4) from thyroid.
- **ACTH (Adrenocorticotropic Hormone):** Stimulates synthesis and secretion of steroid hormones (glucocorticoids) from adrenal cortex.

- **LH (Luteinizing Hormone):** In males, stimulates synthesis/secretion of androgens from testis. In females, induces ovulation of fully mature follicles (Graafian follicles) and maintains corpus luteum.
- **FSH (Follicle Stimulating Hormone):** In males, along with androgens, regulates spermatogenesis. In females, stimulates growth and development of ovarian follicles.
- **MSH:** Acts on melanocytes (pigment-containing cells) and regulates skin pigmentation.
- **Oxytocin:** Acts on smooth muscles. Stimulates vigorous contractions of uterus at child birth and milk ejection from mammary gland.
- **Vasopressin / ADH (Antidiuretic Hormone):** Acts mainly at DCT/collecting duct of kidney to stimulate reabsorption of water and electrolytes, reducing diuresis. Deficiency causes **Diabetes Insipidus** (excessive loss of water in urine $ ightarrow$ dehydration & thirst).

5. PINEAL GLAND

- Located on the **dorsal side of forebrain**.
- Secretes **Melatonin** hormone.
- Functions: Regulates 24-hour diurnal rhythm (sleep-wake cycle, body temperature), influences metabolism, pigmentation, menstrual cycle, and defense capability.

6. THYROID GLAND

Composed of 2 lobes located on either side of trachea, interconnected by a thin flap of connective tissue called **isthmus**. Composed of follicles and stromal tissues.

- **Follicular Cells:** Synthesize **T4 (Tetraiodothyronine / Thyroxine)** and **T3 (Triiodothyronine)**. Iodine is essential for normal rate of hormone synthesis.
- **Hypothyroidism:**
  - During pregnancy: Causes defective development & maturation of fetus leading to **Cretinism** (stunted growth, mental retardation, low IQ, abnormal skin, deaf-mutism).
  - In adult women: Irregular menstrual cycle. Simple goitre (enlargement of thyroid due to iodine deficiency).
- **Hyperthyroidism:** Due to cancer of thyroid or nodules. Causes **Exophthalmic Goitre / Graves' Disease** (characterized by enlarged thyroid, protrusion of eyeballs, increased BMR, weight loss).
- **Thyrocalcitonin (TCT):** Non-iodinated protein hormone secreted by thyroid that **lowers blood calcium levels** (Hypocalcemic).
- **General Functions of T3 & T4:** Regulate basal metabolic rate (BMR), support RBC formation, control metabolism of carbohydrates, proteins & fats, maintain water & electrolyte balance.

7. PARATHYROID GLAND

- Four parathyroid glands present on the **back side of thyroid gland** (one pair each in two lobes).
- Secretes **PTH (Parathormone)** — a peptide hormone. Secretion is regulated by circulating levels of Ca^{2+} ions.
- **Action of PTH: Hypercalcemic hormone** (increases blood Ca^{2+} level).
  - Acts on bones to stimulate bone resorption (demineralization).
  - Reabsorbs Ca^{2+} from renal tubules.
  - Increases Ca^{2+} absorption from digested food.
- **PTH and TCT act antagonistically** to maintain Ca^{2+} homeostasis in body.

8. THYMUS GLAND

- Lobular structure located on **dorsal side of heart and aorta**.
- Plays major role in development of immune system by secreting **Thymosins** (peptide hormones).
- Functions of Thymosins:
  - Differentiation of **T-lymphocytes** (provides Cell-Mediated Immunity).
  - Promotes production of antibodies for **Humoral Immunity**.

- Thymus degenerates in old individuals $ ightarrow$ decreased thymosins $ ightarrow$ weakened immune response in old age.

9. ADRENAL GLAND

One pair located at the anterior part of each kidney. Consists of Adrenal Cortex (outer) and Adrenal Medulla (inner).

A. Adrenal Medulla:

- Secretes 2 hormones: **Epinephrine (Adrenaline)** and **Norepinephrine (Noradrenaline)** — collectively called **Catecholamines**.
- Rapidly secreted in response to stress or emergency $ ightarrow$ Emergency Hormones / Hormones of **Fight or Flight**.
- Effects: Increase alertness, pupillary dilation, piloerection (raising of hairs), sweating, increase heart beat, strength of heart contraction, respiratory rate. Stimulate breakdown of glycogen (glycogenolysis) $ ightarrow$ increased blood glucose; also stimulate breakdown of lipids and proteins.

B. Adrenal Cortex:

Divided into 3 layers: Outer **Zona glomerulosa**, Middle **Zona fasciculata**, Inner **Zona reticularis**.

Group	Main Hormone	Major Functions
Glucocorticoids	Cortisol	• Stimulates gluconeogenesis, lipolysis, proteolysis. • Inhibits cellular uptake and utilization of amino acids. • Cortisol maintains cardio-vascular system and kidney functions. • Anti-inflammatory reactions and suppresses immune response. • Stimulates RBC production.
Mineralocorticoids	Aldosterone	• Acts at renal tubules: stimulates reabsorption of Na^+ and water, excretion of K^+ and phosphate ions. • Maintains electrolytes, body fluid volume, osmotic pressure & blood pressure.
Androgenic Steroids	Small amounts	• Play role in growth of axial hair, pubic hair, and facial hair during puberty.

> ***Disease Note:*** *Underproduction of hormones by adrenal cortex alters carbohydrate metabolism causing acute weakness and fatigue, leading to* ***Addison's Disease****.*

10. PANCREAS

A composite / heterocrine (exocrine + endocrine) gland. Endocrine part consists of 1 to 2 million **Islets of Langerhans** (1-2% of pancreatic tissue).

- **Alpha cells ($ lpha$-cells):** Secrete **Glucagon** (peptide hormone).
  - **Hyperglycemic hormone:** Increases blood glucose levels.
  - Acts on hepatocytes $ ightarrow$ stimulates glycogenolysis & gluconeogenesis. Reducess cellular glucose uptake.
- **Beta cells ($ eta$-cells):** Secrete **Insulin** (peptide hormone).
  - **Hypoglycemic hormone:** Lowers blood glucose levels.
  - Acts on hepatocytes & adipocytes $ ightarrow$ enhances cellular glucose uptake and utilization. Rapid movement of glucose from blood to hepatocytes/adipocytes. Stimulates glycogenesis (conversion of glucose to glycogen).
- **Diabetes Mellitus:** Prolonged hyperglycemia causes diabetes mellitus, characterized by loss of glucose in urine (**Glycosuria**) and formation of harmful ketone bodies (**Ketonuria**). Treated with insulin therapy.

11. GONADS (TESTIS & OVARY)

- **Testis (Male):** Located in scrotum. Dual role as primary sex organ and endocrine gland.
  - **Leydig Cells (Interstitial Cells):** Produce group of hormones called **Androgens** (mainly **Testosterone**).

- Functions: Regulates development, maturation and functions of male accessory sex organs (epididymis, vas deferens, seminal vesicles, prostate, urethra). Stimulates spermatogenesis, male secondary sexual characters, aggressiveness, low pitch voice, anabolic effect on protein/carbohydrate metabolism.

- **Ovary (Female):** Located in abdomen. Primary female sex organ.
  - Produces two groups of steroid hormones: **Estrogen** and **Progesterone**.
  - **Estrogen:** Synthesized by growing ovarian follicles. Controls growth & activity of female secondary sex organs, ovarian follicles development, high pitch voice, mammary gland development.
  - **Progesterone:** Secreted by **Corpus Luteum** (ruptured follicle after ovulation). Supports pregnancy; acts on mammary glands to stimulate formation of alveoli (sac-like structures that store milk) and milk secretion.

12. HORMONES OF HEART, KIDNEY & GASTROINTESTINAL TRACT

Organ / Tissue	Hormone Secreted	Chemical Nature & Action
Heart (Atrial Wall)	ANF (Atrial Natriuretic Factor)	Peptide hormone. When BP increases, ANF is secreted $ightarrow$ causes dilation of blood vessels $ightarrow$ **decreases blood pressure**. (Antagonistic to RAAS).
Kidney (JGA cells)	Erythropoietin	Peptide hormone. Stimulates **Erythropoiesis** (formation of RBCs).
GI Tract	• Gastrin • Secretin • CCK (Cholecystokinin) • GIP (Gastric Inhibitory Peptide)	• **Gastrin:** Stimulates gastric glands to secrete HCl & Pepsinogen. • **Secretin:** Acts on exocrine pancreas $ightarrow$ stimulates water & HCO_3^- ion secretion. • **CCK:** Acts on pancreas & gall bladder $ightarrow$ stimulates secretion of pancreatic enzymes & bile juice. • **GIP:** Inhibits gastric secretion and motility.

13. CLASSIFICATION OF HORMONES BY CHEMICAL NATURE

- **1. Peptide, Polypeptide, Protein Hormones:** Insulin, Glucagon, Pituitary hormones, Hypothalamic hormones, Thymosins, TCT, ANF, Erythropoietin, GI hormones.
- **2. Steroids:** Cortisol, Testosterone, Estradiol, Progesterone, Aldosterone.
- **3. Iodothyronines:** Thyroid hormones (T_3, T_4).
- **4. Amino-Acid Derivatives:** Epinephrine, Norepinephrine (derived from Tyrosine).

14. MECHANISM OF HORMONE ACTION

Hormones produce effects on target tissues by binding to specific proteins called **Hormone Receptors**. Each receptor is specific to one hormone only (high specificity).

A. Water-Soluble Hormones (Protein/Peptide Hormones, Epinephrine):

- Do NOT enter target cell. Bind to **membrane-bound receptors** on cell membrane.
- Form Hormone-Receptor complex $ightarrow$ generates **second messengers** (e.g., **cyclic AMP ($cAMP$), IP_3, Ca^{2+}**).
- Second messengers regulate cellular metabolism $ightarrow$ Biochemical responses $ightarrow$ Physiological responses (e.g., FSH action on ovarian cell).

B. Lipid-Soluble Hormones (Steroid Hormones, Iodothyronines):

- Pass through cell membrane and interact with **intracellular receptors** (mostly nuclear receptors).
- Hormone-Receptor complex interacts with the genome/chromosome $ightarrow$ regulates gene expression / protein synthesis $ightarrow$ Physiological responses (e.g., Estrogen, Cortisol action).